

21).

The Pauli matrices:

$$\sigma_0 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The Dirac matrices:

$$\gamma^0 = \begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix}, \quad \gamma^1 = \begin{pmatrix} 0 & \sigma_1 \\ -\sigma_1 & 0 \end{pmatrix}, \quad \gamma^2 = \begin{pmatrix} 0 & \sigma_2 \\ -\sigma_2 & 0 \end{pmatrix}, \quad \gamma^3 = \begin{pmatrix} 0 & \sigma_3 \\ -\sigma_3 & 0 \end{pmatrix}.$$

The Dirac adjoint of $\psi \in \mathbb{R}^4$

$$\bar{\psi} := \psi^\dagger \gamma^0. \quad \text{row vector}$$

1x4 4x4

for the Dirac field

$$\bar{\psi} := \psi^\dagger \gamma^0. \quad (\psi_1, \psi_2, \psi_3, \psi_4) \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = (\psi_1, \psi_2, -\psi_3, -\psi_4)$$

Let $\mathcal{R} = \mathcal{B} \otimes \mathcal{F} \otimes \mathcal{F}'$


 the fermion Fock space
for electrons and positrons.


 the bosonic Fock space of $\mathcal{H} = L^2(X_0, dx_0, \mathbb{C}^4)$

On $\mathcal{B}$, we defined the electromagnetic four-potential $A = (A_\mu)_{0 \leq \mu \leq 3}$.

QED is described by a Hamiltonian $H = H_0 + H_I$


 natural free Hamiltonian.

Where the interaction Hamiltonian

$$H_I = e \sum_{\mu=0}^3 \int d^3x : \bar{\psi}(0, \mathbf{x}) \gamma^\mu \psi(0, \mathbf{x}) A_\mu(0, \mathbf{x}) : \quad \begin{array}{l} \text{29.1} \\ \text{26.3} \end{array} \quad \text{potential (23.4)}$$

$$e = \sqrt{4\pi\alpha}, \quad \alpha \approx 1/137: \text{ the fine structure constant.}$$

ambiguity: H_I is not defined as an operator on $\mathcal{R}$.

◦ Hamiltonian density:

$$\mathcal{H}(x) = e \sum_{\mu=0}^3 : \bar{\psi}(x) \gamma^\mu \psi(x) A_\mu(x) :$$

21.3 two incoming electrons with momenta p_1, p_2 and spin s_1, s_2 .

Want to compute the probability amplitude of getting
(cf. 21.1)

two outgoing electrons with p_3, p_4, s_3, s_4 .

◦ the scattering amplitude:

$$\underbrace{\langle 0 | a(p_3, s_3) a(p_4, s_4)}_{\text{outgoing state}} S \underbrace{a^\dagger(p_1, s_1) a^\dagger(p_2, s_2) | 0 \rangle}_{\text{incoming state}}$$

where S is the scattering operator.

◦ The Dyson expansion: as 14.3

Suppose p_1, p_2, p_3, p_4 : distinct.

the n th term :

$$\frac{(-i)^n}{n!} \int \dots \int d^4x_1 \dots d^4x_n \langle 0 | a(p_3, s_3) a(p_4, s_4) \mathcal{T} \mathcal{H}(x_1) \mathcal{H}(x_2) \dots \mathcal{H}(x_n) a^\dagger(p_1, s_1) a^\dagger(p_2, s_2) | 0 \rangle$$

Which is a linear combination of many terms of the form

$$\langle 0 | a(p_3, s_3) a(p_4, s_4) \mathcal{T} (: \bar{\psi}_{j_1}(x_1) \psi_{k_1}(x_1) A_{\mu_1}(x_1) ; \dots ; \bar{\psi}_{j_n}(x_n) \psi_{k_n}(x_n) A_{\mu_n}(x_n) :) a^\dagger(p_1, s_1) a^\dagger(p_2, s_2) | 0 \rangle$$

• The Feynman diagrams for QED. (p. 96)

18.2 Recall

p. 93, $\langle 0 | B_j B_k | 0 \rangle$ in Wick theorem is called a contraction of B_j and B_k .

$$S = 1 - \frac{ig}{4!} \int d^4x : \varphi(x)^4 : + \mathcal{O}(g^2). \quad : \text{first order}$$

$$\begin{aligned} \langle p_3, p_4 | S | p_1, p_2 \rangle &= - \frac{ig}{4!} \int d^4x \underbrace{\langle p_3, p_4 | : \varphi(x)^4 : | p_1, p_2 \rangle}_{= \langle 0 | a(p_3) a(p_4) : \varphi(x)^4 : a^\dagger(p_1) a^\dagger(p_2) | 0 \rangle} + \mathcal{O}(g^2). \end{aligned}$$

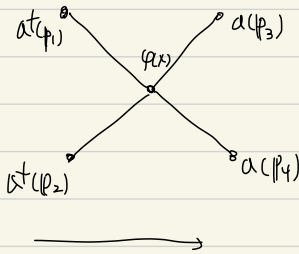
By the Wick theorem, we consider, for example,

$$(*) \quad \langle 0 | a(p_3) \varphi(x) | 0 \rangle \langle 0 | a(p_4) \varphi(x) | 0 \rangle \langle 0 | \varphi(x) a^\dagger(p_1) | 0 \rangle \langle 0 | \varphi(x) a^\dagger(p_2) | 0 \rangle.$$

Each product of nonzero contractions corresponds uniquely to a graph with some additional structure called a Feynman diagram. ([1] p. 155.)

This is rough description. Please, see the other reference. [1] ch.6.

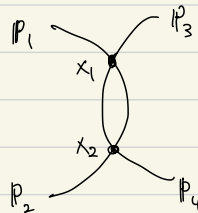
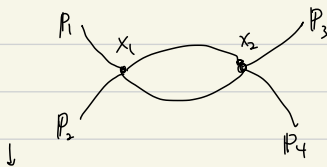
- put the vertices for creation operators at the left, the vertices for annihilation operators at the right, and the internal vertices in between.
- For each contraction there is an edge of graph btw the approximate vertices.
- Each line is associated to a particular particle species, namely, the one to which the operators in its contraction pertain. — $\sim$...
- The lines of the graph associated to particles with distinct antiparticle are equipped with arrows under some rules.
(like photon, no arrows are drawn on lines or to particle without a distinct antiparticle.)



⊗ Second order

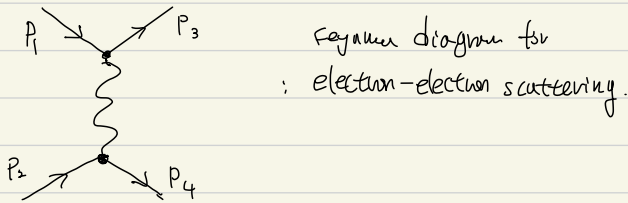
→ Wick ordering: there cannot be self-loops at the x_i 's

$$\langle 0 | a(p_3) a(p_4) \overline{ \left(: \varphi(x_1)^4 : ; : \varphi(x_2)^4 : \right) } a^\dagger(p_1) a^\dagger(p_2) | 0 \rangle$$



$$\begin{aligned} & \langle 0 | \varphi(x_1) a^\dagger(p_1) | 0 \rangle \langle 0 | \varphi(x_2) a^\dagger(p_2) | 0 \rangle \langle 0 | \overline{ \varphi(x_1) \varphi(x_2) } | 0 \rangle^2 \\ & \times \langle 0 | a(p_3) \varphi(x_2) | 0 \rangle \langle 0 | a(p_4) \varphi(x_2) | 0 \rangle. \end{aligned}$$

Now for QED the basic ideas are the same, but the algebra is more complicated.



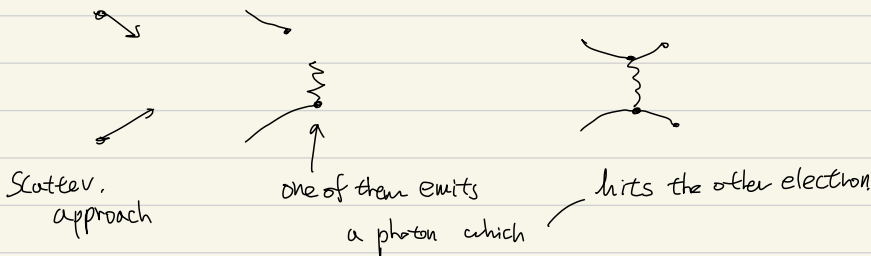
There is the momentum-space integral for this diagram.
And the position-space integration over that gives from

$$\frac{1}{i^2} \iint \langle 0 | a(p_3, s_3) a(p_4, s_4) A_{\mu}(x_1) \bar{\psi}_1(x_1) \gamma^{\mu} \psi(x_1) A_{\nu}(x_2) \bar{\psi}(x_2) \gamma^{\nu} \psi(x_2) a^{\dagger}(p_2, s_2) a^{\dagger}(p_1, s_1) | 0 \rangle d^4x d^4y$$

by pairing up the operators into the contractions

Notation for contractions
(See [1])

$$\frac{1}{i^2} \iint \overbrace{a(p_3, s_3) \bar{\psi}_1(x_1)} \gamma^{\mu} \overbrace{\psi(x_1) a^{\dagger}(p_1, s_1)} \overbrace{A_{\mu}(x_1) A_{\nu}(x_2)} \times \overbrace{a(p_4, s_4) \bar{\psi}(x_2)} \gamma^{\nu} \overbrace{\psi(x_2) a^{\dagger}(p_2, s_2)} d^4x d^4y.$$



Lec. 28. cf. [1] ch. 6.5

We are trying to compute the second order term in Dyson expansion for the scattering amplitude

$$\langle 0 | a(p_3, s_3) a(p_4, s_4) \int a^\dagger(p_1, s_1) a^\dagger(p_2, s_2) | 0 \rangle.$$

Recall that the Feynman propagator for a particle of mass m is given for

Def 9.1 (ch. 9.2)

$$\Delta_{F,m}(x) = \lim_{\varepsilon \rightarrow 0} \int \frac{d^4 p}{(2\pi)^4} \frac{e^{-i(x,p)}}{-p^2 + m^2 - i\varepsilon}$$

↑
emphasis for mass.

: a fundamental solution of the Klein-Gordon equation:
 $(\partial_t^2 - \nabla_x^2 + m^2) \Delta_F = \delta$

The Dirac propagator is a 4×4 array of Schwartz distributions defined as:

$$\Delta_D(x) = \sum_{\mu=0}^3 (i \gamma^\mu \partial_\mu + m) \Delta_{F,m}$$

: a fundamental solution of the Dirac operator:
 $(-i \gamma^\mu \partial_\mu + m) \Delta_D = \delta.$

This appears in QED calculation: For any $1 \leq j, k \leq 4$,

$$\langle 0 | T \psi_j(x_1) \psi_k(x_2) | 0 \rangle = - \langle 0 | T \psi_j(x_1) \psi_k(x_2) | 0 \rangle = -i (\Delta_D(x_1 - x_2))_{jk}. \quad (\text{cf. Proposition 9.1})$$

appear in higher orders of perturbation theory.

The photon propagator $\Delta_{\mu\nu}$, for $0 \leq \mu, \nu \leq 3$, is defined as
(Feynman gauge)

$$\Delta_{\mu\nu} = -\eta_{\mu\nu} \Delta_{F,0}(x) \quad \eta = \text{diag}(1, -1, -1, -1)$$

$$\langle 0 | T A_\mu(x_1) A_\nu(x_2) | 0 \rangle = -i \Delta_{\mu\nu}(x_1 - x_2).$$

28.3

the second order term in the Dyson expansion of the electron scattering amplitude

$$\langle 0 | u(p_3, s_3) a(p_4, s_4) S a^\dagger(p_1, s_1) a^\dagger(p_2, s_2) | 0 \rangle$$
 is

$$\sum_{\mu, \nu=0}^3 \frac{im^2 e^2}{\sqrt{16} \omega_{p_1} \omega_{p_2} \omega_{p_3} \omega_{p_4}} \bar{u}(p_3, s_3) \gamma^\mu u(p_1, s_1) \frac{2\eta_{\mu\nu}}{(p_3 - p_2)^2} \bar{u}(p_4, s_4) \gamma^\nu u(p_2, s_2) + \text{similar terms. } (*)$$

28.4

the non-relativistic limit of the scattering amplitude; $|p_i| \ll 1$ for each i .

$$\bullet \omega_{p_i} \approx m \text{ for each } i.$$

$$\left. \begin{aligned} (p_3 - p_1)^2 &= (p_3^0 - p_1^0)^2 - (p_3 - p_1)^2 \\ p_i^0 &= \sqrt{m^2 + p_i^2} \approx m + \frac{p_i^2}{2m} \end{aligned} \right\} \Rightarrow (p_3 - p_1)^2 = (p_3 - p_1)^2 (1 + \alpha(1))$$

$$\bullet \bar{u}(p, s) \gamma^\mu u(p', s') \approx \bar{u}(0, s) \gamma^\mu u(0, s')$$

$$\Rightarrow \sum_{\mu} \bar{u}(0, s) \gamma^\mu u(0, s') = 2 \delta_{s, s'}, \quad \bar{u} = u^\dagger \gamma^0, \quad u(0, s) = \underset{p=(m, 0)}{f_p} f_s = f_s.$$

$$\odot \bar{u}(0, s) \gamma^0 u(0, s') = 2 \delta_{s, s'}$$

$$(1 \ 0 \ 0 \ 0) \begin{pmatrix} \sigma_0 & 0 \\ 0 & -\sigma_0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = (1 \ 0 \ 1 \ 0) \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \xrightarrow{\text{in case 1}} = 0, \quad \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix} = 2$$

$$\bar{u}(0, s) \gamma^j u(0, s') = 0 \text{ for } j=1, 2, 3,$$

(11)

Thus

$$(*) \approx \frac{-ie^2}{(p_3 - p_1)^2} \delta_{s_1, s_3} \delta_{s_2, s_4} (2\pi)^4 \delta^{(4)}(p_3 + p_4 - p_1 - p_2) + \text{similar terms.}$$

On the other hand, the function $V: \mathbb{R}^3 \rightarrow \mathbb{R}$ whose Fourier transform is

$$\hat{V}(p) = \frac{1}{p^2}$$

is the function

$$V(x) = \frac{1}{4\pi|x|}.$$

Concentrate on the first particle.

Recall the Born approximation in p.62,

$$\langle p_3 | V | p_1 \rangle = \hat{V}(p_3 - p_1) \approx (*)$$

$\rightarrow V$ is the Coulomb potential.

Coulomb potential arises naturally in the non-relativistic limit of QED.

and we recover the classical physics in the low-energy limit.

28.5 A successes of QED.

The magnetic moment of the electron differ from the $\frac{e}{2m}$ predicted by the Dirac equation by the factor $1 + F_2(0)$, which is the one-loop approximation is

$$1 + F_2(0) = 1 + \frac{e^2}{8\pi^2} \approx 1.0011614$$

$\uparrow$ so-called anomalous magnetic moment of the electron
(Schwinger in 1949)

Lec. 29.

29.1 Interacting Quantum fields.

• Interacting quantum fields : Important object and Complicated than free fields.

• $\mathcal{H} = L^2(X_m, d\mu_m)$ and the scalar free field φ , say φ^{free} .

the Hamiltonian $H = H_0 + gH_I$, where H_0 : Hamiltonian of free evolution,

$$H_I = \frac{1}{4!} \int d^3x : \varphi^{\text{free}}(0, \mathbf{x})^4 :$$

The interacting field φ defined by H is

$$\varphi(t, \mathbf{x}) := e^{itH} \varphi^{\text{free}}(0, \mathbf{x}) e^{-itH}. \quad = \text{does not make mathematical sense.}$$

Note $\varphi(0, \mathbf{x}) = \varphi^{\text{free}}(0, \mathbf{x})$.

(rigorous definition is an open problem)

$$\varphi^{\text{free}}(t, \mathbf{x}) = e^{itH_0} \varphi^{\text{free}}(0, \mathbf{x}) e^{-itH_0}$$

29.3 The Wightman axioms.

[1] In the late 1950s, Arthur Wightman, a mathematical physicist, and Lars Gröding, a mathematician on PDE who was a thesis advisor for Lars Hörmander,

formulated a list of basic properties that any physically reasonable and mathematically well-defined quantum field theory should have. This list has come to be known as the Gröding-Wightman axioms.

Only consider the axioms for bosonic scalar fields.

(A1) There is a Hilbert space $\mathcal{H}$ whose elements represent the possible states of the quantum system, and a unitary rep. U of $\mathcal{P}$ in $\mathcal{H}$.

• $(U(a_i))_{a \in \mathbb{R}^{1,3}}$ is a commuting family of unitary operators.

• $E(A) = 0$ whenever $A \cap \{p \in \mathbb{R}^{1,3} : p^0 \geq |p|\} = \emptyset$.

there is no particles with "negative masses".

(A2) $\exists \mathcal{D} \subset \mathcal{H}$ dense, $\exists \varphi: S(\mathbb{R}^4) \rightarrow \{\text{w.b.d. operators on } \mathcal{H}\}$: linear

such that for $f \in S(\mathbb{R}^4)$, $\varphi(f)$ is defined on $\mathcal{D}$ and $\varphi(f)(\mathcal{D}) \subset \mathcal{D}$.

$$(\varphi(f)\xi, \eta) = (\xi, \varphi(f)\eta) \text{ for } \xi, \eta \in \mathcal{D}.$$

(A3) $\forall \xi, \eta \in \mathbb{D}, f \mapsto (\xi, \eta f)$: a tempered distribution.

(A4) $\forall (a, A) \in \mathcal{P},$

$$U(a, A) \varphi(f) U(a, A)^{-1} = \varphi(V(a, A)f),$$

where

$$V(a, A)f(x) = f(A^{-1}(x-a)).$$

ensure compatibility with special relativity.

(A5) $f, g \in \mathcal{S}(\mathbb{R}^4)$. If $\sup_{\mathbb{R}^4} f, \sup_{\mathbb{R}^4} g$ are "spacelike separated", then

$$[\phi(f), \phi(g)] = 0$$

ensure compatibility with the stipulation of special relativity that a signal cannot travel faster than the speed of light.

(A6) $\exists! \Omega \in \mathbb{D}$ such that $U(a, 1)\Omega = \Omega$ for any a .

ensure the existence and uniqueness of the vacuum state.

(A7) The closed linear span of elements like $\varphi(t_1) \dots \varphi(t_n)\Omega$ for $n, t_1, \dots, t_n$ arbitrary, is the Hilbert space $\mathcal{H}$.

ensure that the field ϕ is "rich enough" to generate all possible states.

Under this axiom, one can prove the spin-statistics theorem that particles of integer spin must be bosons and particles of half-integer spin must be fermions, and the PCT theorem etc.

Unfortunately, attempts to produce examples (of field theories that satisfy the Wightman axioms and have nontrivial interaction) in physical space-time $\mathbb{R}^4$ have yet to succeed.